MEDISCAN AI

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Clinical Analysis Report

Analyzed on AMD MI300X · 192GB VRAM · ROCm 7.2 · Qwen2-VL-7B

Name: **avi**
Age: **45**
Gender: **Male**
Blood Type: **A-**
Smoking: **No**
Alcohol: **No**
Family Hist: **No**

Image: X-Ray
Region: Knee and Proximal Lower Leg
Quality: Clear

CANCER RISK ASSESSMENT

CRITICAL

STAGE: NONE

95
Score / 100

Suspected Type: **Osteosarcoma (or other primary aggressive bone malignancy)**

The radiographic features of the tibial lesion are highly characteristic of an aggressive primary malignant bone tumor, such as osteosarcoma. Key features include the expansile nature, mixed lytic and sclerotic bone destruction, cortical breach, and prominent spiculated periosteal reaction. These findings, combined with the patient's severe leg pain, indicate a critical risk of malignancy.

KEY FINDINGS

1. Bone Lesion: A large, expansile, and highly aggressive lesion is identified arising from the proximal metadiaphysis of the tibia. It demonstrates a mixed lytic and sclerotic pattern with significant cortical destruction, a broad zone of transition, and prominent spiculated periosteal reaction, suggestive of a 'sunburst' appearance. There is likely an associated soft tissue mass component. This appearance is highly suggestive of an aggressive primary bone malignancy.
(Critical)

DIFFERENTIAL DIAGNOSIS

Suspected Osteosarcoma (Primary Bone Malignancy)

Severe Leg Pain (symptom likely due to bone lesion)

HIGH Confidence. The classic radiographic signs of an aggressive bone lesion with mixed lytic/sclerotic components, cortical destruction, and spiculated periosteal reaction strongly point towards a primary bone malignancy, with osteosarcoma being a leading differential diagnosis.

HIGH Confidence. The large and aggressive bone lesion in the proximal tibia is the likely etiology of the patient's reported severe leg pain.

RECOMMENDED NEXT STEPS

1. **Urgent MRI of the affected limb** — Urgent
   To precisely delineate the intraosseous and extraosseous extent of the tumor, assess marrow involvement, and evaluate neurovascular bundles, which is critical for surgical planning.

2. **Image-guided Biopsy of the tibial lesion** — Urgent
   To obtain tissue for histopathological diagnosis, which is essential for confirming the type of malignancy and guiding treatment.

3. **Staging studies (e.g., CT chest, PET-CT or Bone Scan)** — Urgent
   To evaluate for regional and distant metastatic disease, as aggressive bone tumors have a high potential for spread.

4. **Consultation with an Orthopedic Oncologist** — Urgent
   Specialized expertise is required for the comprehensive management, including surgical and oncological treatment, of primary bone malignancies.

VALIDATION

Agent 3 — Validator

Reliability: 8/10

VERDICT: Pass with Warning

The medical findings, severity assessment, and diagnostic confidence are highly consistent and well-supported by the described imaging features, indicating a medically sound interpretation. The suggested next steps are appropriate and urgent given the critical nature of the suspected condition. However, a critical general medical disclaimer is absent, which is a safety concern for a report intended for user consumption.

Missing explicit general medical disclaimer reminding users to consult a healthcare professional and stating that the report does not constitute medical advice.

MediScan AI is an assistive AI tool. Reports are not a substitute for professional medical diagnosis. Always consult a licensed physician.